

Chapter 1

LASER

1.1 Introduction

LASER is an acronym for Light Amplification by Stimulated Emission of Radiation. The first LASER was built by Theodore H Maiman in the year 1960. Thus it finds various applications starting from industries to communication.

1.2 Characteristics of a LASER beam

The LASER beam has the following four characteristics.

1. LASER beam is highly monochromatic.
2. LASER beam is highly coherent.
3. LASER beam is highly directional.
4. LASER is a high intensity beam of light.

1.3 Interaction of radiation with matter

The interaction between radiation and matter occurs through the following three processes.

1. Induced absorption
2. Spontaneous emission
3. Stimulated emission

1.3.1 Induced Absorption

When a photon of right energy is incident on the atom then the photon is absorbed. This process is induced by the photon and hence it is called Induced Absorption.

Consider an atom in a lower energy state E_1 , it will excite to higher energy state E_2 by absorbing the incident photon of energy $E = h\nu = E_2 - E_1$. Here E_1 energy of the lower energy state, E_2 is the energy of the higher energy state, h is the Planck's constant ν is the frequency of photon.

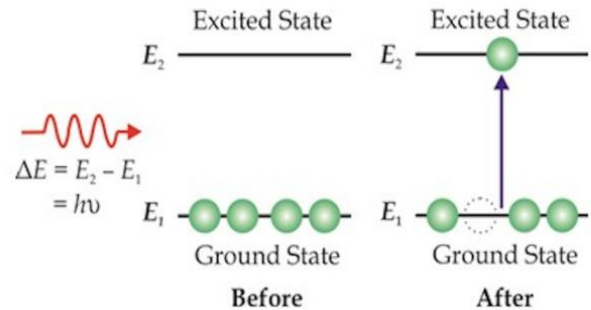


Figure 1.1: Induced absorption

1.3.2 Spontaneous Emission

Spontaneous emission is the process of emission of photon, when an atom transits from higher energy level to lower energy level without the influence of any external energy.

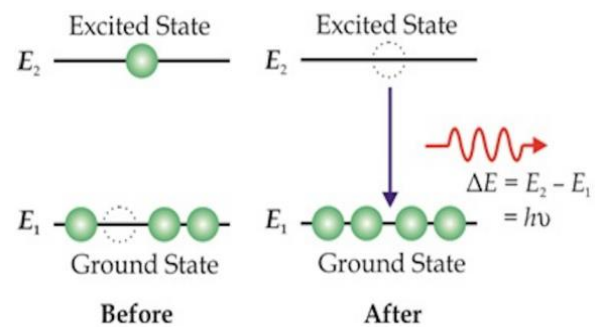


Figure 1.2: Spontaneous emission

An electron in the higher energy state of an atom makes a transition to lower energy state without the action of any external agency. The energy of the photon emitted is given by $E = h\nu = E_2 - E_1$. In this process the emitted photons need not travel in the same direction. Thus the light beam is not directional.

1.3.3 Stimulated Emission

When a photon of suitable energy interacts with an atom in the higher energy state then the atom is stimulated (Forced) to make transition from higher energy state to a lower energy state with the emission of a photon. Both the incident photon and the emitted photons are coherent and travel in the same direction. Thus the process is called stimulated emission.

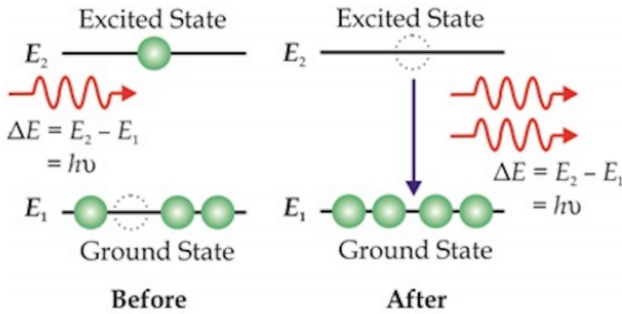


Figure 1.3: Stimulated emission

When a photon of energy $h\nu = E_2 - E_1$ interacts with an atom in the higher energy state the stimulated emission takes place with the emission of two photons of same energy that are highly directional and coherent. Thus stimulated emission could be used to generate a highly coherent directional beam of light.

1.4 Einstein's A and B co-efficients and expression for energy density

Consider a system containing N atoms and is under thermal equilibrium. Let E_1 and E_2 be the lower and higher energy levels that contain N_1 and N_2 number of atoms respectively. Let the incident energy density of the radiation be E_ν . Hence the system absorbs and emits the energy through the following processes. The energy of the photons absorbed and emitted by the atoms is $E = h\nu = (E_2 - E_1)$

Rate of induced absorption

The rate of induced absorption is defined as the number of induced absorption per second per unit volume in unit time. Rate of absorption depends on

1. Number of atoms in the lower energy state N_1 .
2. The incident energy density E_ν .

Hence

1. Rate of Induced absorption $\propto N_1 E_\nu$

2. Rate of Induced absorption $= B_{12} N_1 E_\nu$

Here B_{12} is proportionality constant called Einsteins coefficient of Induced absorption.

Rate of spontaneous emission:

The number of spontaneous emission per unit volume in unit time is called rate of spontaneous emission. Rate of spontaneous emission depends on

Since spontaneous emission is a voluntary process it is independent of energy density E_ν . The rate of spontaneous emission depends only on the number of atoms in the higher energy state N_2 . Thus

1. Rate of spontaneous emission $\propto N_2$
2. Rate of Spontaneous emission $= A_{21} N_2$

Here A_{21} is the proportionality constant called Einstein's co-efficient of spontaneous emission.

Rate of stimulated emission

The number of stimulated emission per unit volume in unit time is called rate of stimulated emission. Rate of stimulated emission depends upon,

1. Number of atoms in the higher energy state (N_2)
2. The energy density (E_ν).

Hence

1. The Rate of stimulated emission $\propto N_2 E_\nu$
2. Rate of stimulated emission $= B_{21} N_2 E_\nu$

Here the proportionality constant called B_{21} is Einstein's coefficient of stimulated emission.

Under Thermal Equilibrium the total Energy of the System remains unchanged. Hence Rate of Absorption is equal to rate of emission.

$\therefore$ Rate of Induced Absorption = [Rate of Spontaneous emission + Rate of Stimulated Emission]

$\therefore$

$$B_{12} N_1 E_\nu = A_{21} N_2 + B_{21} N_2 E_\nu \quad (1.1)$$

$$(B_{12} N_1 - B_{21} N_2) E_\nu = A_{21} N_2$$

$$E_\nu = \frac{A_{21} N_2}{B_{12} N_1 - B_{21} N_2} \quad (1.2)$$

$$E_\nu = \frac{A_{21}}{B_{12} \frac{N_1}{N_2} - B_{21}} \quad (1.3)$$

$$E_\nu = \frac{A_{21}}{B_{21}} \left[\frac{1}{\frac{B_{12}}{B_{21}} \frac{N_1}{N_2} - 1} \right] \quad (1.4)$$

According to Boltzmann relation the we have

$$\frac{N_2}{N_1} = e^{\frac{-h\nu}{kT}} \quad (1.5)$$

or we can re-write as,

$$\frac{N_1}{N_2} = e^{\frac{h\nu}{kT}} \quad (1.6)$$

Here h is the Planck's constant, c is the speed of light in vacuum, λ is the wavelength of the photon, k is the Boltzmann constant and T is the absolute temperature. Substituting for $\frac{N_1}{N_2}$ in equation 1.4

$$E_\nu = \frac{A_{21}}{B_{21}} \left[\frac{1}{\frac{B_{12}}{B_{21}} \left(e^{\frac{h\nu}{kT}} \right) - 1} \right] \quad (1.7)$$

According to Planck's radiation law, the equation for energy density in the frequency domain is given by

$$E_\nu = \frac{8\pi h\nu^3}{c^3} \left[\frac{1}{e^{\frac{h\nu}{kT}} - 1} \right] \quad (1.8)$$

on comparing equations 1.7 and 1.8 we can get

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} \quad (1.9)$$

and

$$\frac{B_{12}}{B_{21}} = 1 \quad (1.10)$$

or $B_{12} = B_{21}$

This means that Probability of Induced absorption is equal to Probability of Stimulated emission. Hence A_{21} & B_{21} can be replaced by A & B . Thus equation 1.7 could be written as

$$E_\nu = \frac{A}{B} \left[\frac{1}{e^{\frac{h\nu}{kT}} - 1} \right] \quad (1.11)$$

Hence the expression for energy density in terms of Einstein's co-efficient A and B .

1.5 LASER Action and the Conditions for LASER action

Consider a LASER system. Let an atom in the excited state is stimulated by a photon of right energy so that atom makes stimulated emission. Two coherent photons are obtained. These two coherent photons if stimulate two atoms in the excited state to make emission then four coherent photons are emitted. These four coherent photons stimulate 4 more atoms in the excited state resulting in 8 coherent photons and so on. As the process continues number of coherent photons increases. These coherent photons constitute an intense beam of LASER. This phenomenon of building up of number of coherent photons so as to get an intense LASER beam is called lasing action.

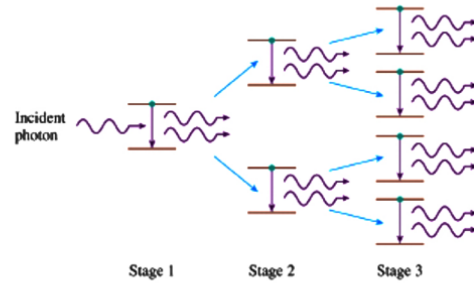


Figure 1.4: LASER action

LASER action could be achieved through the conditions population inversion and meta-stable state.

1.5.1 Population Inversion and Pumping

If a system is under thermal equilibrium the number of atoms in excited state is less than the number of atoms in the lower energy state. For the production of LASER number of stimulated emission must be more when compared to induced absorption and spontaneous emission. This is possible only if the number of atoms in the higher energy state is more than the number of atoms in the lower energy state and is called population inversion. The means of achieving population inversion by supplying energy from a suitable source is called Pumping. In addition, to have more stimulated emissions, the life time of atoms in the excited state must be longer. Thus the essential conditions for population inversion are

1. Higher energy state should possess a longer life time.
2. The number of atoms in the higher energy state must be greater than the number of atoms in the lower energy state.

1.5.2 meta-stable state

The life time of an energy level is of the order of 10^{-8} second. If an atom possesses unusual longer life time in an energy state such a state is referred to as a meta-stable state. Usually the life time of meta-stable state varies from 10^{-2} s to 10^{-3} s. Population inversion could be achieved with the help of three energy states with one of them a meta-stable state and is as shown in the figure 1.5. The population inversion is achieved between the state E_2 and E_1 as state E_2 is a meta-stable state.

Note : The principles of Laser are

1. Stimulated Emission
2. Population Inversion
3. meta-stable State

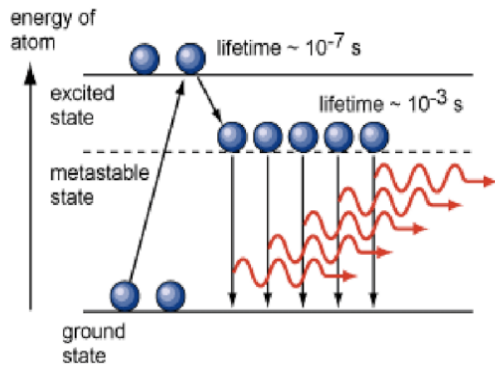


Figure 1.5: Population inversion, E_2 is meta-stable state

1.6 Requisites of a LASER system

The three requisites of a LASER system are,

1. Excitation source for pumping action
2. Active medium that supports meta-stable states
3. LASER cavity

1.6.1 Energy Source or Pumping Mechanism

In order to achieve population inversion more and more atoms are to be moved to higher energy state and is called pumping. This is achieved by supplying suitable energy using an energy source. If optical energy is used then the pumping is called optical pumping and if electrical energy is used then the pumping is called electrical pumping.

1.6.2 Active medium

Population inversion occurs at certain stage in the active medium due to the absorption of energy. The active medium supports meta-stable states. After this stage the active medium is capable of emitting LASER light.

1.6.3 Resonant cavity (or) LASER cavity

The LASER Cavity consists of an active medium bound between two highly parallel mirrors. The reflection of photons from the mirrors results in multiple traverse of photons through the active medium inducing more and more stimulated emissions. Thus amplification of light is achieved. This also helps to tap certain permissible part of LASER energy from the active medium. The cavity resonates and the output will be maximum when the distance L between the mirrors is equal to an integral multiple of $\frac{\lambda}{2}$. Here λ is the wavelength of incident suitable radiation. The length of the LASER cavity is expressed as

$$L = \frac{n\lambda}{2} \quad (1.12)$$

1.7 Semiconductor LASER or Diode LASER

Introduction

Semiconductor diode LASER is an LED with heavily doped P and N sections. First semiconductor LASER was fabricated in 1962 using $Ga - As$ by Hall with his co-workers. It is a low cost and high efficiency LASER.

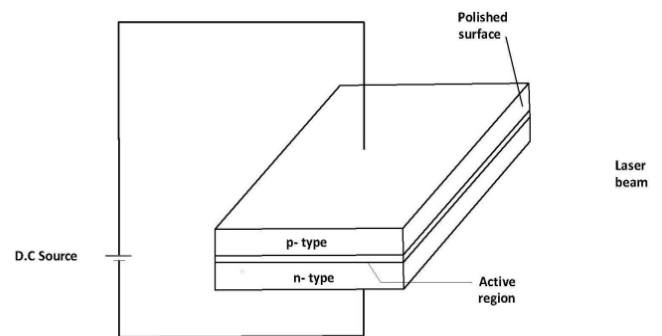
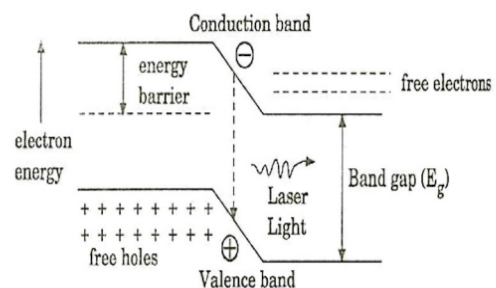


Figure 1.6: Semiconductor Diode LASER

Construction

Construction

The $Ga - As$ LASER diode belongs to direct band gap semiconductors. The n -section is derived by doping the substrate with *Tellurium* and p -section is derived by doping the substrate with *Zinc*. The diode used is in the form of a cube with dimension 0.4 mm. The depletion region is of thickness 0.1 micrometer and lies horizontal as shown in the figure 1.6. The current is passed through the ohmic contacts provided to the top and bottom faces. The front and back faces are polished and made highly parallel to each other to have a LASER cavity. The other two faces are roughened.



Working

The Diode is forward biased using an external source. Therefore electrons and holes flow across junction. Hence

the current flows through the diode. When a hole meets an electron it recombines with electron emitting a photon. This could be considered as the transition of electron from conduction band to valance band. When the current is low spontaneous emission is predominant. If the current is sufficiently high population inversion is achieved. The photons liberated initially due to spontaneous emissions induce further stimulated emissions. The LASER cavity helps in the amplification of light. Finally this results in an avalanche of photons and hence the LASER action is achieved. If the GaAs semiconductor is used then the wavelength of the LASER emitted is 840nm.

Advantages

1. It has excellent efficiency
2. The output can be modulated
3. It produces both continuous wave output or pulsed output.
4. It is highly economical

Applications

1. It is used in optical fiber communication.
2. It is used in commercial CD recording and reading.
3. It is used in Barcode Reader, Laser printing and Laser Cooling.

Optical Modulators

Optical modulators are devices that control the properties of light such as amplitude, phase, or polarization, usually by applying an external electric field. Two important mechanisms are based on electro-optic effects: the Pockels effect and the Kerr effect.

Pockels Effect (Linear Electro-Optic Effect)

Certain crystals, such as Lithium Niobate (LiNbO_3) and Potassium Dihydrogen Phosphate (KDP), exhibit a change in refractive index that is **linearly proportional** to the applied electric field.

When polarized light passes through such a crystal under an applied voltage, its phase changes. With the help of polarizers, this phase modulation can be converted into amplitude modulation, allowing the light to be switched on or off.

Applications:

- High-speed switching in optical communication.
- Optical shutters in laser experiments.

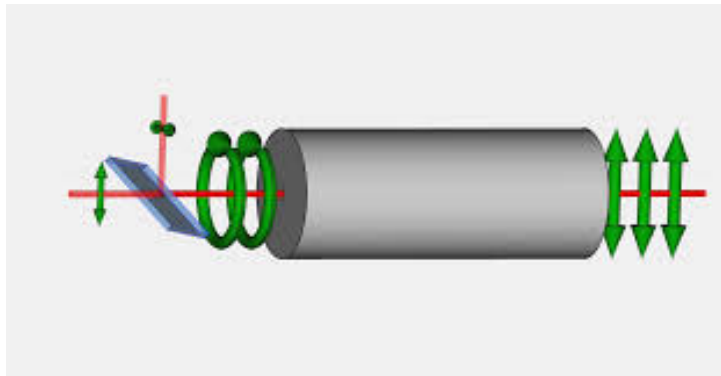


Figure 3: Working of a Pockels cell as an electro-optic modulator.

Kerr Effect (Quadratic Electro-Optic Effect)

In certain materials, the refractive index changes **quadratically** with the applied electric field. This phenomenon, called the Kerr effect, is weaker than the Pockels effect but can occur in symmetric materials where the Pockels effect is absent. A Kerr cell can rotate the polarization of light depending on the voltage applied.

Applications:

- Ultrafast optical switching in laser systems.
- Phase and amplitude modulation in optical communication.

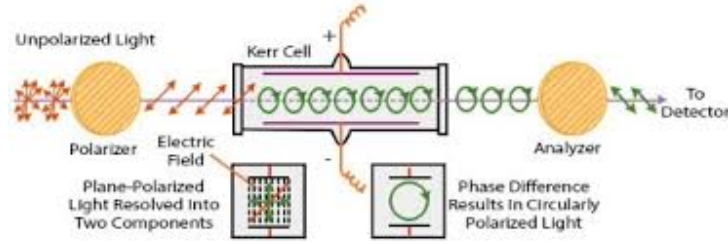


Figure 4: Illustration of the Kerr effect in an optical medium.

Photodetectors for Single-Photon Detection

Photodetectors are devices that convert single photons into measurable electrical signals. Two widely used technologies are Single Photon Avalanche Diodes (SPADs) and Superconducting Nanowire Single-Photon Detectors (SNSPDs).

Single Photon Avalanche Diode (SPAD)

A SPAD is a semiconductor diode operated in reverse-bias mode above its breakdown voltage (also known as Geiger mode). When a single photon enters, it creates an electron-hole pair. This triggers an avalanche multiplication process, producing a measurable electrical pulse.

Features:

- Compact and operates at room temperature.
- High detection efficiency (30–70%).
- Higher dark count rates compared to superconducting detectors.

Applications:

- Quantum cryptography.
- Fluorescence lifetime imaging.
- LiDAR and single-molecule detection.

Superconducting Nanowire Single-Photon Detector (SNSPD)

An SNSPD consists of a thin superconducting wire cooled close to absolute zero. When a photon hits the nanowire, it creates a localized **hotspot** that disrupts superconductivity, forming a resistive region. This produces a measurable voltage pulse that signifies the arrival of the photon.

Features:

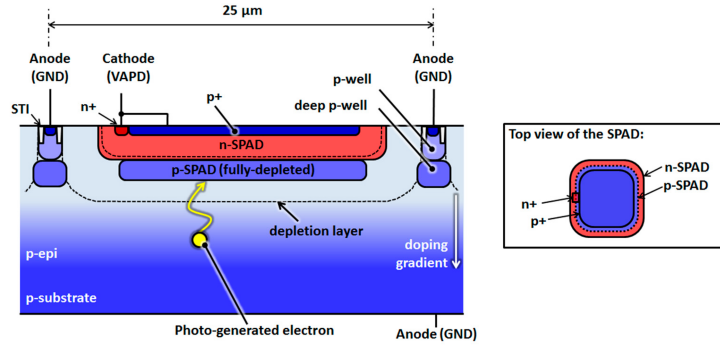


Figure 5: Detection mechanism of a Single Photon Avalanche Diode (SPAD).

- Extremely high detection efficiency (greater than 90%).
- Very low dark counts (false signals).
- Excellent timing resolution (on the order of picoseconds).
- Requires cryogenic cooling.

Applications:

- Quantum key distribution (QKD).
- Quantum computing experiments.
- Astronomy (detecting faint light from distant sources).

Conclusion

Attenuators are essential for preparing weak photon sources, optical modulators such as Pockels and Kerr cells allow control of light, and photodetectors such as SPADs and SNSPDs enable the measurement of single photons. Together, these devices form the foundation of modern quantum optics, quantum communication, and quantum computing systems.

Mach-Zehnder Interferometer

Principle

The Mach-Zehnder Interferometer (MZI) works on the principle of interference of light waves. A beam of coherent light is split into two beams using a beam splitter. These two beams travel along different paths, where they accumulate a phase difference due to differences in path lengths or refractive index variations. The beams are then recombined at another beam splitter. Depending on the relative phase shift, the recombined beams

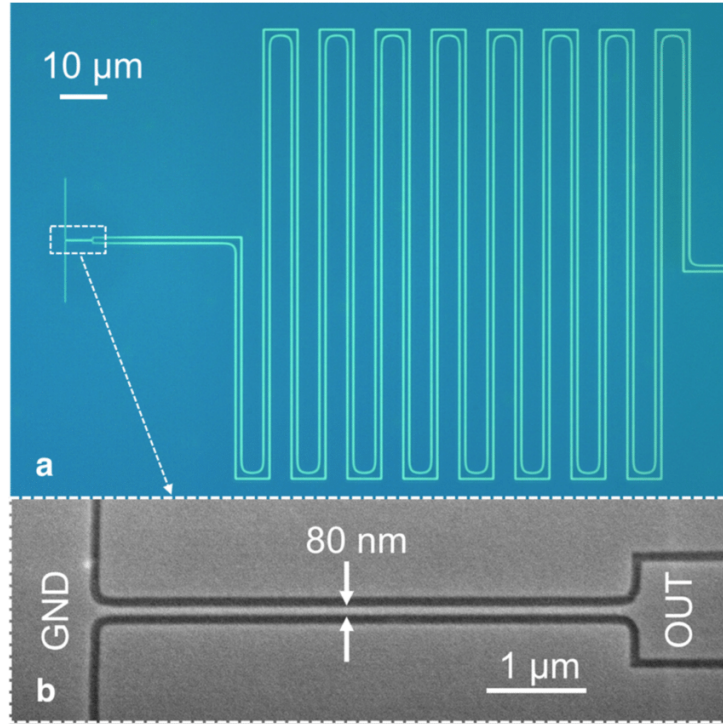


Figure 6: Working principle of a Superconducting Nanowire Single-Photon Detector (SNSPD).

interfere constructively (bright output) or destructively (dark output). Thus, the output intensity depends directly on the phase difference between the two arms.

Construction

The main components of an MZI are:

1. **Light Source:** A coherent source such as a laser.
2. **First Beam Splitter (BS1):** Splits the incoming light into two beams.
3. **Mirrors (M1, M2):** Redirect the two beams along different paths.
4. **Phase Modifying Medium (optional):** Introduces an additional phase shift in one arm.
5. **Second Beam Splitter (BS2):** Recombines the beams.
6. **Detectors (D1, D2):** Detect the output intensities and interference pattern.

Working

1. The input light enters BS1 and splits into two beams.
2. Each beam travels along a separate path, reflecting off mirrors M1 and M2.

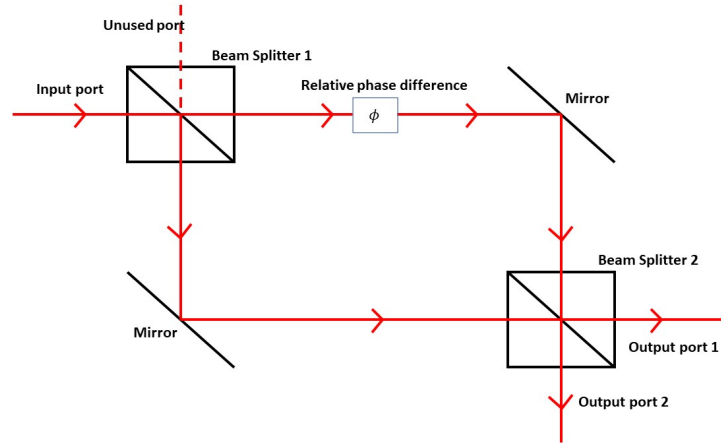


Figure 7: Schematic diagram of a Mach–Zehnder Interferometer.

3. If one path is longer or contains a medium with higher refractive index, the beam acquires an additional phase shift.
4. At BS2, the two beams recombine.
5. Depending on the phase difference:
 - **Constructive Interference:** Phase difference is an even multiple of π , giving maximum intensity.
 - **Destructive Interference:** Phase difference is an odd multiple of π , giving minimum intensity.

By analyzing the output intensity, one can measure extremely small phase changes due to path length variations or medium properties.

Applications

- Measuring refractive index changes in materials.
- Optical sensing and biosensing.
- Quantum optics and single-photon interference experiments.
- Used as an electro-optic modulator (via Pockels effect).
- Telecommunications: modulation and signal encoding in fiber optic systems.

Chapter 2

Optical Fibers

2.1 Introduction

Optical fibers are the wires and strands made of transparent dielectrics which guide light over longer distances using the phenomenon of **Total Internal Reflection**. Many optical fibers are bundled together and are given a protective layer of covering using an insulating material. This bundle is called Optical Fiber Cable or Fiber Bundle (Bundle Fiber).

Construction: The sectional view of a typical optical fiber is as shown in the figure. It has three regions named Core, Cladding and Sheath.

1. The innermost light guiding region is called Core.
2. The layer covering core and helps in total internal reflection of light is called Cladding or Clad.
3. The outermost protective layer is called Sheath (Coating). The sheath protects the fiber from mechanical stress and chemical reactions.

The optical fiber is designed to support total internal reflection and hence the RI of core n_1 is made greater than the RI of cladding n_2 . A typical fiber will be of the order of few microns.

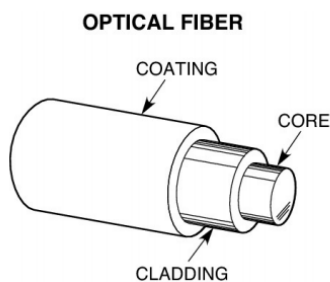


Figure 2.1: Optical fiber construction

2.2 Total Internal Reflection

Consider a ray of light moving from a denser medium to a rarer medium. As a result the incident ray of light bends away from the normal. Hence the angle of refraction is greater than the angle of incidence. As the angle of incidence increases the angle of refraction also increases. For a particular angle of incidence θ_c the refracted ray grazes the interface separating the two media. The corresponding angle of incidence θ_c is called Critical Angle. If the angle of incidence is greater than the critical angle then all the light is turned back into the same medium and is called Total Internal Reflection.

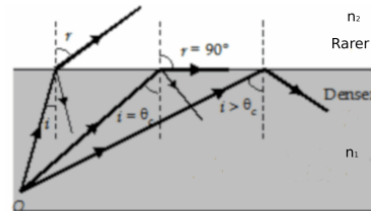


Figure 2.2: Total Internal Reflection

According to Snell's law

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\text{when } \theta_1 = \theta_c \text{ then, } \theta_2 = 90^\circ$$

$$n_1 \sin \theta_c = n_2 \sin 90^\circ$$

$$\sin \theta_c = \frac{n_2}{n_1}$$

$$\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right) \quad (2.1)$$

2.3 Angle of acceptance and Numerical aperture

Acceptance angle (θ_0) is the maximum angle of incidence with which the ray is sent into the fiber core which allows

the incident light to be guided by the core. It is also called as waveguide acceptance angle or acceptance cone half angle.

In optics, the numerical aperture (NA) of an optical fiber is a dimensionless number that characterizes the range of angles over which the fiber can accept light. Numerical aperture represents the light gathering capability of optical fiber and it is given by $NA = \sin\theta_0$.

2.3.1 Condition for propagation

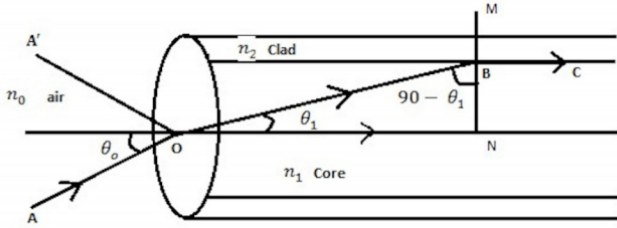


Figure 2.3: Ray propagation in the fiber

Consider an optical fiber with core made of refractive index n_1 & cladding made of material refractive index n_2 . Let n_0 be the refractive index of the surrounding medium. Let a ray of light AO entering into core at an angle of incidence θ_0 w.r.t fiber axis. Then it is refracted along OB at an angle θ_1 & meet core-cladding interface at critical angle of incidence ($\theta_c = 90 - \theta_1$). Then the refracted ray grazes along BC . On applying Snell's law at O , we get

$$\begin{aligned} n_0 \sin\theta_0 &= n_1 \sin\theta_1 \\ \therefore \sin\theta_0 &= \frac{n_1}{n_0} \sin\theta_1 \end{aligned} \quad (2.2)$$

On applying Snell's law at point B , we get

$$\begin{aligned} n_1 \sin(90^\circ - \theta_1) &= n_2 \sin 90^\circ \\ n_1 \cos\theta_1 &= n_2 \\ \therefore \cos\theta_1 &= \frac{n_2}{n_1} \end{aligned} \quad (2.3)$$

From trigonometric identity

$$\begin{aligned} \sin^2\theta_1 + \cos^2\theta_1 &= 1 \\ \sin\theta_1 &= \sqrt{1 - \cos^2\theta_1} \end{aligned}$$

using equation 1.25

$$\begin{aligned} \sin\theta_1 &= \sqrt{1 - \left(\frac{n_2}{n_1}\right)^2} \\ \sin\theta_1 &= \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}} \end{aligned}$$

$$\sin\theta_1 = \frac{1}{n_1} \sqrt{n_1^2 - n_2^2} \quad (2.4)$$

use equation (1.26) in equation (1.24) we have,

$$\begin{aligned} \sin\theta_0 &= \frac{n_1}{n_0} \frac{1}{n_1} \sqrt{n_1^2 - n_2^2} \\ \sin\theta_0 &= \frac{1}{n_0} \sqrt{n_1^2 - n_2^2} \end{aligned} \quad (2.5)$$

Numerical aperture $N.A = \sin\theta_0$

$$N.A = \frac{1}{n_0} \sqrt{n_1^2 - n_2^2} \quad (2.6)$$

If the fiber is in air $n_0 = 1$ then,

$$N.A = \sin\theta_0 = \sqrt{n_1^2 - n_2^2} \quad (2.7)$$

Light is transmitted through the fiber only when

$$\theta_i \leq \theta_0 \quad (2.8)$$

$$\sin\theta_i \leq \sin\theta_0 \quad (2.9)$$

$$\sin\theta_i \leq \sqrt{n_1^2 - n_2^2} \quad (2.10)$$

$$\sin\theta_i \leq N.A \quad (2.11)$$

This is the condition for propagation. Light will be transmitted through the optical fiber with multiple total internal reflections when the above condition is satisfied.

2.4 Modes of propagation

Though optical fiber should support any numbers of rays for propagation practically. But it is found that the optical fiber allows only a certain restricted number of rays for propagation. The maximum number of rays or paths supported by the fiber for the propagation of light is called *Modes of propagation*. Based on the modes of propagation fibers are classified into Single mode and Multi-mode fibers.

2.5 RI Profile

The RI profile is a plot of variation of RI of the fiber with respect to radial distance from the axis of an optical fiber. Based on the RI profile fibers are classified into Step index and Graded index fibers. In case of Step index fibers RI of the core is constant. In case of Graded index fibers the RI decreases radially outwards.

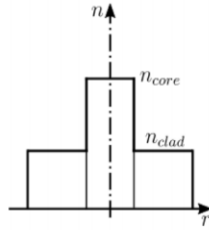


Figure 2.4: RI profile of step index fiber

2.5.1 Types of optical fibers

In any optical fiber, the whole material of the cladding has a uniform refractive index value. But the refractive index of the core material may either remain constant or subjected to variation in a particular pattern. The curve which represents the variation of refractive index with respect the radial distance from the axis of the fiber is called the *refractive index profile*. The optical fibers are classified under 3 categories,

1. Step index single mode fiber
2. Step index multi-mode fiber
3. Graded index multi-mode fiber

This classification is done depending on the refractive index profile, and the number of modes that the fiber can guide.

Step index single mode fiber

A single mode step index fiber consists of a very fine thin core of uniform RI surrounded by Cladding of RI lower than that of Core. Since there is abrupt change in the RI of Core and Cladding at the interface it is called step index fiber. Since the Core size is small the Numerical aperture is also small and hence support single mode. They accept light from LASER source. Splicing is difficult. They are used in submarine cables.

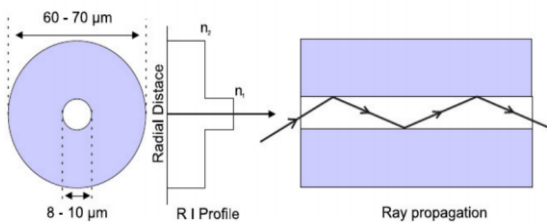


Figure 2.5: Step index single mode fiber

Step index multi-mode fiber

This is similar to single mode step index fiber with the exception that it has a larger core diameter. The core diameter

is very large as compared to single mode optical fiber. A typical multi-mode step index fiber is as shown in figure. The numerical aperture is large because of large core size and thus support multi-modes. They accept light from both LASER as well as from LED. They are used in data links.

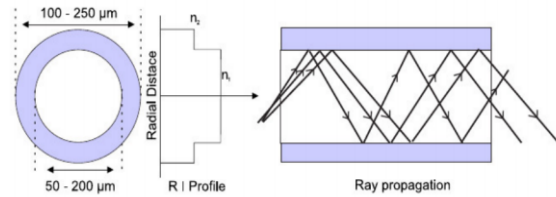


Figure 2.6: Step index multi-mode fiber

Graded index multi-mode fiber

A multi-mode fiber has concentric layers of RI is called GRIN fiber. It means the R I of the Core varies with distance from the fiber axis. The RI is maximum at the center and decreases with radial distance towards to core-cladding interface. The R I profile is as shown in fig. In GRIN fibers the acceptance angle and numerical aperture diminish with radial distance. The light transmission is as shown above. They accept light from both LASER as well as from LED. They are used for medium distance communication for example telephone link between central offices.

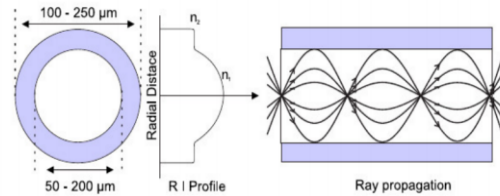


Figure 2.7: Graded index multimode fiber

2.6 Attenuation

The optical energy (signal) passing through the optical fiber gets reduced progressively. This is due to attenuation. It is also called the fiber loss or significant loss. The attenuation is measured in terms of attenuation co-efficient. The attenuation co-efficient α is defined as the ratio of optical power output to the optical power input for a fiber of length L and for a given wavelength of propagating light. It is expressed in dB/km . Attenuation co-efficient is given by

$$\alpha = \frac{-10}{L} \log_{10} \left(\frac{P_{out}}{P_{in}} \right) \quad dB/km \quad (2.12)$$

Here L is the length of the cable in km , P_{in} is Power of optical signal at launching end (input power) & P_{out} is Power of optical signal at receiving end (output power)

The attenuation in fibers gives is due to the following three losses

1. Absorption losses
2. Scattering loss (due to Rayleigh Scattering)
3. Geometric Effects (Radiation losses)

2.6.1 Absorption loss

In this type of loss, the loss of signal power occurs due to absorption of photons associated with the signal. Photons are absorbed either by impurities in the glass fiber or by pure glass material itself. Absorption loss is wavelength dependent. Thus absorption loss is classified in to two types.

Extrinsic absorption : Extrinsic loss in an optical fiber is due to the absorption of light by the impurities such as hydroxide ions and transition metal ions such as iron, chromium, cobalt and copper.

Intrinsic absorption Intrinsic loss in fiber is due to the absorption of light by the material of the fiber glass itself. The intrinsic losses are insignificant.

2.6.2 Scattering loss

Light traveling through the core can get scattered by impurities or small regions with sudden change in refractive index. Rayleigh scattering varies as $\alpha = \frac{1}{\lambda^4}$ and leads to significant power loss at smaller wavelengths. The scattering results in loss of photons. Rayleigh scattering is responsible for maximum losses in optical fibers.

2.6.3 Geometric effects

These may occur due to manufacturing defects like irregularities in fiber dimensions during drawing process or during coating, cabling or insulation processes. The microscopic bends are the bends with radii greater than fiber diameter. The microscopic bends couple light between the various guided modes of the fiber and some of them then leak through the fiber.

2.7 Applications of Optical Fibers

2.7.1 Fiber Optic Networking

Local Area Network

A Local Area Network (LAN) is a type of computer network that interconnects multiple computers and computer-

driven devices in a particular physical location. Traditionally copper coaxial cables are used for LAN.

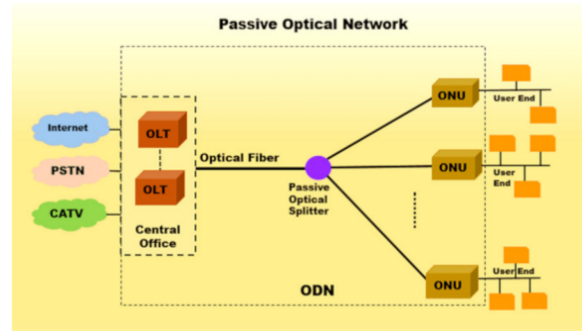


Figure 2.8: Fiber Optic LAN

Abbreviations

1. PON - Passive Optical Network
2. ONT - Optical Network Terminal
3. ODN - Optical Distribution Network
4. OLT - Optical Line Terminal
5. ONU - Optical Network Unit

Passive Optical LAN

A passive optical network refers to a fiber-optic network utilizing a point-to-multipoint topology and optical splitters to deliver data from a single transmission point to multiple user endpoints. Passive here refers to the unpowered condition of the fiber and splitting/combining components. Passive optical LANs are built entirely using Optical fiber cables. The passive optical LAN works on the concept of optical network terminals (ONT) and passive optical splitters. Network switches act as passive splitters and the commercial media converters act as optical network terminals in a real-time application of passive optical LAN.

Advantages

1. High speeds and bandwidth
2. Longer distances are possible
3. Less chance of errors

2.7.2 Point to point communication using Optical Fibers

In an optical fiber communication system, the input signals (audio, video or other digital data) are used to modulate light from a source like a LED or a semiconductor LASER and is transmitted through optical fiber. At the